

CASCADE-CLASSICAL CORRESPONDENCE III: REFINEMENT SPACES AND SYMMETRY ACTIONS

ABSTRACT. This paper builds the finite-level analytic infrastructure intended for downstream cascade-correspondence papers. From the algebraic substrate of paper P2, we construct: iterated refinement data $\mathcal{R}_n^\bullet = (V_n^\bullet, E_n^\bullet, F_n^\bullet, \partial_n^\bullet)$ for $\bullet \in \{H_4, D_4\}$, including explicit 2-face data needed to place centroid vertices, with literal vertex inclusion $V(G_n^\bullet) \subset V(G_{n+1}^\bullet)$; finite-level ℓ^2 vertex spaces X_n^0 and edge ℓ^2 spaces X_n^1 under counting measure; downward bonding maps $p_{n+1,n}^0$ defined as restriction to $V(G_n^\bullet)$ and $p_{n+1,n}^1$ defined as fibrewise averaging over the canonical edge-parent map; the projective inverse-limit Hilbert space (in the vertex sector) $X_{\text{inv}}^{0,\bullet} \cong \ell^2(V_\infty^\bullet; F^0)$; the cylindrical inductive-limit space $X_{\text{cyl}}^{0,\bullet} = c_{00}(V_\infty^\bullet; F^0)$ embedded densely in $X_{\text{inv}}^{0,\bullet}$; full intertwining identities for the refinement-Coxeter and refinement- σ pairs; and a restricted intertwining identity for the Coxeter- σ pair (full $W(H_4)$ - σ commutation on the V_{H_4} sector fails because $W(H_4)$ acts via $\mathbb{Q}(\varphi)$ -linear matrices; see Theorem 8.5(4) and Remark 8.6 for the precise scope).

This is the third of thirteen planned cascade infrastructure papers. The construction is in the correct projective direction: bonding maps $p_{n+1,n}$ go *downward* from level $n+1$ to level n , implemented as restriction (vertex sector) or as canonical parent-fibre averaging (edge sector). Their adjoints $j_{n,n+1} := p_{n+1,n}^*$ are (rescaled) embeddings going upward. In the vertex sector the adjoint is a genuine section ($p^0 j^0 = \text{id}$), the cylindrical direct limit is well-defined, and the inverse limit is its Hilbert completion. In the edge sector the adjoint is only a *half-section* ($p^1 j^1 = \frac{1}{2} \text{id}$), a clean cylindrical direct-limit / Hilbert-completion package is not established here, and edge-sector statements are kept at the finite level with bounded bonding maps only (Remarks 6.3 and 7.5). *No continuum-limit, Gromov-Hausdorff, spectral-convergence, or manifold-identification claim is made here.*

The main theorems proved in this paper: (1) Proposition 2.8, the refinement rule with $V_n \subset V_{n+1}$ inclusion is well-defined and Coxeter-equivariant; (2) Proposition 4.3, each finite-level X_n^0, X_n^1 is a finite-dimensional Hilbert space; (3) Theorem 5.3, downward bonding maps are contractions under counting measure; (4) Proposition 5.5, the cylindrical embeddings $j_{n,n+1}$ are the adjoints of the bonding maps; (5) Theorem 6.2, the vertex algebraic direct limit $X_{\text{cyl}}^{0,\bullet}$ is well-defined as $c_{00}(V_\infty^\bullet; F^0)$; (6) Theorem 7.3, the vertex inverse limit $X_{\text{inv}}^{0,\bullet}$ under restriction is canonically isometric to $\ell^2(V_\infty^\bullet; F^0)$ and is a separable Hilbert space; (7) Proposition 7.4, $X_{\text{cyl}}^{0,\bullet}$ embeds densely in $X_{\text{inv}}^{0,\bullet}$; (8) Theorem 8.5, the commutation $g \circ p_{n+1,n} = p_{n+1,n} \circ g$ for g in the relevant Coxeter group, and analogously for σ from P1.

These are the load-bearing inputs for the closure-dynamics paper P4, the metric projection paper P5, and the gauge/arithmetic/computation branch papers P9-P13, as well as for the correspondence Foundations paper.

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1. INTRODUCTION AND SOURCE AUDIT

1.1. The repair target. An earlier draft of the correspondence Foundations manuscript defined the cascade limit Hilbert space using forward zero-extension maps $X_n \rightarrow X_{n+1}$ and called the resulting object an “inverse limit”. That construction is incorrect on two counts: the system is inductive (not projective), and forward zero-extension is not isometric under any naturally-normalized ℓ^2 structure on finite refinement levels.

The correct construction is built on *downward* bonding maps $p_{n+1,n}: X_{n+1} \rightarrow X_n$. The cleanest mathematical choice for these bonding maps, used here, is:

- Vertex sector: $p_{n+1,n}^0$ is restriction of a function on $V(G_{n+1}^\bullet)$ to its sub-domain $V(G_n^\bullet) \subset V(G_{n+1}^\bullet)$.
- Edge sector: $p_{n+1,n}^1$ is fibrewise averaging along the canonical edge-parent map $E^{\text{or}}(G_{n+1}^\bullet) \rightarrow E^{\text{or}}(G_n^\bullet)$, which assigns each subdivided level- $(n+1)$ oriented edge to the level- n oriented edge it is a piece of.

Both maps are genuine contractions under *counting measure* on finite vertex/edge sets, and their adjoints are the natural embeddings in the upward direction. The projective system in the correct direction is then $X_{\text{inv}} := \{(f_n) : p_{n+1,n}f_{n+1} = f_n\}$, equipped with the limit inner product, and we show this is canonically isometric to $\ell^2(V_\infty^\bullet)$ for the countable union $V_\infty^\bullet := \bigcup_n V(G_n^\bullet)$.

1.2. What this paper proves. The substrate constructed here consists of:

- (1) Iterated refinement graphs $G_n^{H_4}$ and $G_n^{D_4}$ with $V(G_n^\bullet) \subset V(G_{n+1}^\bullet)$ and a canonical edge-parent map (§2).
- (2) Counting measures on vertices and oriented edges (§3).
- (3) Finite-level vertex and edge ℓ^2 spaces X_n^0, X_n^1 , each a finite-dimensional Hilbert space (§4).
- (4) Downward bonding maps $p_{n+1,n}^0$ (restriction) and $p_{n+1,n}^1$ (parent-fibre averaging), both contractions (§5).
- (5) Cylindrical embeddings $j_{n,n+1}^0$ (zero-extension) and $j_{n,n+1}^1$ (rescaled replication) as adjoints of the bonding maps (§5).
- (6) Algebraic direct-limit space (vertex sector) $X_{\text{cyl}}^{0,\bullet} \cong c_{00}(V_\infty^\bullet; F^0)$ (§6).

- (7) Projective inverse-limit Hilbert space (vertex sector) $X_{\text{inv}}^{0,\bullet} \cong \ell^2(V_\infty^\bullet; F^0)$ (§7).
- (8) Commuting refinement / Coxeter / σ intertwining identities for $p_{n+1,n}$ and $j_{n,n+1}$ (§8).

1.3. What this paper does not prove. P3 is deliberately narrow. We record the exclusions explicitly:

- (1) No continuum-limit, Gromov–Hausdorff, or spectral-convergence statement.
- (2) No closure-functional, gradient-flow, energy, or monotonicity theorem.
- (3) No identification of X_{inv} with any classical continuum function space such as $L^2(S^3)$.
- (4) No metric, hydrodynamic, gauge, spectral, zeta, Hecke, or CTM/TM correspondence theorem.
- (5) No vertex-wise quotient map $V(G_n^{H_4}) \rightarrow V(G_n^{D_4})$.
- (6) No physical or phenomenological interpretation.

1.4. Source audit. Theorem inputs to this paper are:

Internal cascade infrastructure.: Paper P1 (cascade–classical correspondence I, σ -fixed rationality [1]) for the coefficientwise σ -action; paper P2 (cascade algebraic substrate [2]) for the algebraic substrate, including $D_4 \subset E_8$, the icosian model of H_4 , the explicit V_{600} and V_{24} vertex sets, and the $\text{Cl}(1,3)$ and octonionic sector models.

Classical mathematics.: Standard Hilbert space and direct/inverse limit theory [4, 5]; standard Coxeter group theory [6, 7]; finite graph and 1-cochain conventions [8]; elementary ℓ^2 theory.

We do not cite as theorem-grade input any other cascade-internal note. The bibliography below contains only the sources actually used in proofs.

1.5. Organization. Section 2 sets up the refinement graphs and the canonical edge-parent map. Section 3 fixes the counting measures. Section 4 defines the finite-level vertex and edge ℓ^2 spaces. Section 5 constructs the downward bonding maps and their adjoints, the cylindrical embeddings. Section 6 treats the cylindrical algebraic direct-limit space. Section 7 constructs the projective limit Hilbert space and proves it is $\ell^2(V_\infty)$. Section 8 establishes the intertwining identities for refinement, Coxeter, and σ . Section 9 states the citation contract for downstream papers. Appendix A is a notation concordance.

2. REFINEMENT GRAPHS AND THE EDGE-PARENT MAP

2.1. The base graphs.

Definition 2.1 (Base graphs). Let $V_{600} \subset \mathbb{H} \cong \mathbb{R}^4$ be the standard 120-vertex set of the regular 600-cell, in the coordinate realization of [2, §5]. Let $V_{24} \subset V_{600}$ be the 24-vertex subset (axis points and half-type points) congruent to the rescaled D_4 root polytope by [2, Prop. 5.3].

The *base H_4 graph* $G_0^{H_4} = (V_0^{H_4}, E_0^{H_4})$ has vertex set $V_0^{H_4} := V_{600}$ and unoriented edges joining pairs of 600-cell vertices at the standard 600-cell edge distance [7, §22.7].

The *base D_4 graph* $G_0^{D_4} = (V_0^{D_4}, E_0^{D_4})$ has vertex set $V_0^{D_4} := V_{24}$ and unoriented edges joining pairs of 24-cell vertices at the standard 24-cell edge distance [7, §22.2].

We write $\bullet$ for either decoration and $G_n^\bullet$ for the level- n graph below. The Coxeter group $W^\bullet$ is $W(H_4)$ when $\bullet = H_4$ and $W(D_4)$ when $\bullet = D_4$.

2.2. Refinement with vertex inclusion. A graph alone does not determine the Schläfli refinement; the construction needs explicit 2-face data, since centroids are indexed by 2-faces. We carry that data through levels as a *refinement datum*.

Definition 2.2 (Refinement datum). A *refinement datum at level n* is a tuple $\mathcal{R}_n^\bullet = (V_n^\bullet, E_n^\bullet, F_n^\bullet, \partial_n^\bullet)$ where $V_n^\bullet$ and $E_n^\bullet$ are the vertex and unoriented edge sets of $G_n^\bullet$; $F_n^\bullet$ is a finite set of 2-faces (each face is a finite cyclic sequence of edges $(e_1, \dots, e_k) \in E_n^\bullet$ with $k \geq 3$ closing up into a polygon, with the standard cyclic-equivalence convention); and $\partial_n^\bullet: F_n^\bullet \rightarrow \bigsqcup_k (E_n^\bullet)^k$ is the boundary map sending each face to its edge sequence. The base data $\mathcal{R}_0^\bullet$ is induced by Definition 2.1 from the 1-skeleton and 2-faces of the underlying 4-polytope (V_{600} for $\bullet = H_4$, V_{24} for $\bullet = D_4$). $W^\bullet$ acts on each component of $\mathcal{R}_n^\bullet$ compatibly with the boundary map.

Definition 2.3 (Schläfli refinement with vertex inclusion). Given a refinement datum $\mathcal{R}_n^\bullet = (V_n^\bullet, E_n^\bullet, F_n^\bullet, \partial_n^\bullet)$, the *Schläfli refinement* produces a new datum $\mathcal{R}_{n+1}^\bullet$ with the following explicit components:

- **Vertices.** $V_{n+1}^\bullet := V_n^\bullet \sqcup M_n^\bullet \sqcup C_n^\bullet$, where $M_n^\bullet$ contains exactly one edge-midpoint m_e per $e \in E_n^\bullet$, and $C_n^\bullet$ contains exactly one face-centroid c_f per $f \in F_n^\bullet$. The first summand is the literal inclusion $V_n^\bullet \subset V_{n+1}^\bullet$.
- **Edges.** $E_{n+1}^\bullet$ is the disjoint union of three explicit edge classes:
 - (a) *Subdividing edges:* for each $e = \{u, v\} \in E_n^\bullet$, the two edges $\{u, m_e\}$ and $\{m_e, v\}$.
 - (b) *Midpoint-centroid edges:* for each face $f \in F_n^\bullet$ with edge sequence $(e_1, \dots, e_k) = \partial_n^\bullet(f)$, the k edges $\{m_{e_i}, c_f\}$ for $i = 1, \dots, k$.
 - (c) *Centroid-midpoint reconnection edges:* the empty set in this paper. (Some Schläfli-style refinement schemes in the literature add edges between centroids of adjacent faces; we do not include those, and downstream uses of this paper do not require them.)
- **Faces.** $F_{n+1}^\bullet$ is determined by the new edge set: each level- n face f with edge sequence $(e_1, \dots, e_k)$ produces k level- $(n+1)$ triangle faces $(m_{e_i}, c_f, m_{e_{i+1}})$ (cyclic indices), each with edge sequence $(\{m_{e_i}, c_f\}, \{c_f, m_{e_{i+1}}\}, \{m_{e_{i+1}}, m_{e_i}\})$. Note: the third edge of each triangle, $\{m_{e_{i+1}}, m_{e_i}\}$, is *not* an element of $E_{n+1}^\bullet$ as constructed above; we record this as a defect that the present paper does not need (the load-bearing downstream uses of $G_{n+1}^\bullet$ in this paper, and in [?, ?], depend only on the vertex inclusion and the edge classes (a)+(b), not on the level- $(n+1)$ face structure).
- **Boundary map.** $\partial_{n+1}^\bullet$ is induced on the level- $(n+1)$ triangles by the explicit construction above.
- **$W^\bullet$ -action.** $W^\bullet$ acts on $V_{n+1}^\bullet$ by $g \cdot v = gv$ for $v \in V_n^\bullet$, $g \cdot m_e = m_{ge}$ for $m_e \in M_n^\bullet$ (well-defined since $g\{u, v\} = \{gu, gv\}$), and $g \cdot c_f = c_{gf}$ for $c_f \in C_n^\bullet$ (well-defined since $W^\bullet$ acts on $F_n^\bullet$). The induced action on $E_{n+1}^\bullet$ permutes each of the edge classes (a)+(b) among themselves.

Remark 2.4 (Scope of the construction). This paper uses Definition 2.3 only to construct the chain spaces $X_n^{0,\bullet}$, $X_n^{1,\bullet}$, the bonding maps, and the intertwining identities below. The level- $(n+1)$ face structure (and the missing midpoint-midpoint edges) play no role in those constructions. Downstream papers needing a fuller cell-complex structure on $\mathcal{R}_n^\bullet$ may extend Definition 2.3 explicitly; we do not.

Remark 2.5 (Vertex inclusion is the key). The literal subset $V_n^\bullet \subset V_{n+1}^\bullet$ from Definition 2.3 is the structural feature that makes the entire bonding-map construction below work cleanly: restriction of a function on $V_{n+1}^\bullet$ to $V_n^\bullet$ is a well-defined function on $V_n^\bullet$, and (under counting measure) is automatically a contraction in the ℓ^2 -norm.

2.3. The canonical edge-parent map. Unlike vertices, edges are not preserved under refinement: each level- n edge $\{u, v\} \in E_n^\bullet$ is replaced by two level- $(n+1)$ edges $\{u, m\}$ and $\{m, v\}$. We record the inverse direction as a function:

Definition 2.6 (Edge-parent map). For oriented edges, define

$$\pi_{n+1,n}^E: E^{\text{or}}(G_{n+1}^\bullet) \rightarrow E^{\text{or}}(G_n^\bullet) \cup \{\perp\}$$

by:

- If $e' \in E^{\text{or}}(G_{n+1}^\bullet)$ is one of the two pieces of a subdivided level- n oriented edge $e = (u, v) \in E^{\text{or}}(G_n^\bullet)$ — i.e., $e' = (u, m)$ or $e' = (m, v)$ where m is the midpoint of $\{u, v\}$ — then $\pi_{n+1,n}^E(e') := e$ (with the orientation inherited).
- If e' is one of the new level- $(n+1)$ edges connecting a midpoint to a centroid or two midpoints (i.e., a “boundary” edge introduced by Schläfli refinement that does not subdivide a level- n edge), then $\pi_{n+1,n}^E(e') := \perp$.

The edge-parent map is well-defined as a function and is $W^\bullet$ -equivariant: $g \cdot \pi_{n+1,n}^E(e') = \pi_{n+1,n}^E(g \cdot e')$ for every $g \in W^\bullet$ and every $e' \in E^{\text{or}}(G_{n+1}^\bullet)$. This holds because Schläfli refinement is itself $W^\bullet$ -equivariant, so $W^\bullet$ permutes the descendants of an edge among themselves.

Notation 2.7 (Subdividing edges). We write $E_{\text{sub}}^{\text{or}}(G_{n+1}^\bullet) := \{e' : \pi_{n+1,n}^E(e') \neq \perp\}$ for the set of level- $(n+1)$ oriented edges that subdivide a level- n edge. By construction, $|E_{\text{sub}}^{\text{or}}(G_{n+1}^\bullet)| = 2 \cdot |E^{\text{or}}(G_n^\bullet)|$ (each level- n edge contributes exactly two level- $(n+1)$ subdividing edges).

2.4. Properties of the refinement.

Proposition 2.8 (Refinement properties). *For every $n \geq 0$ and $\bullet \in \{H_4, D_4\}$:*

- (1) $V_n^\bullet \subset V_{n+1}^\bullet$ is a literal subset inclusion of finite sets.
- (2) The action of $W^\bullet$ on $V_n^\bullet$ extends to an action on $V_{n+1}^\bullet$ that commutes with the inclusion.
- (3) The edge-parent map $\pi_{n+1,n}^E$ is well-defined and $W^\bullet$ -equivariant.
- (4) Each level- n oriented edge has exactly two level- $(n+1)$ preimages under $\pi_{n+1,n}^E$ (its two subdividing pieces in the inherited orientation).

Proof. (1) and (2) follow directly from Definition 2.3: the vertex inclusion is by construction, and the $W^\bullet$ -action extends because midpoints and centroids are $W^\bullet$ -equivariantly defined from the level- n data.

(3) Edge-parent equivariance: by the Schläfli refinement construction [7, §8.2], $W^\bullet$ permutes subdivided edges among themselves, and the parent-edge assignment respects this permutation.

(4) Each level- n unoriented edge $\{u, v\}$ contributes oriented edges (u, v) and (v, u) at level n , each of which has two subdividing pieces at level $n+1$ (namely $(u, m), (m, v)$ for (u, v) , and $(v, m), (m, u)$ for (v, u)). So there are two preimages per oriented edge, giving the $|E_{\text{sub}}^{\text{or}}(G_{n+1}^\bullet)| = 2|E^{\text{or}}(G_n^\bullet)|$ count of Notation 2.7. $\square$

Remark 2.9 (No non-canonical choices). Definition 2.6 introduces no non-canonical choice: each level- $(n+1)$ oriented edge is either intrinsically a subdividing piece of a unique level- n oriented edge (in which case it has a unique parent), or it is intrinsically a boundary edge of the Schläfli refinement (in which case it has parent $\perp$). No lex-order, evaluation, or vertex-choice rule is needed.

3. COUNTING MEASURES

Definition 3.1 (Counting measures). For each $n \geq 0$ and $\bullet \in \{H_4, D_4\}$:

$$\begin{aligned} \mu_n^\bullet(S) &:= |S|, & S &\subseteq V(G_n^\bullet); \\ \nu_n^\bullet(T) &:= |T|, & T &\subseteq E^{\text{or}}(G_n^\bullet). \end{aligned}$$

These are unweighted counting measures; in particular, every vertex and oriented edge has measure 1.

Remark 3.2 (Why counting measure). We use unscaled counting measure rather than probability normalization. The reason is that, with $V_n^\bullet \subset V_{n+1}^\bullet$, restriction to $V_n^\bullet$ is a contraction in $\ell^2(V_{n+1}^\bullet, \mu_{n+1}^\bullet) \rightarrow \ell^2(V_n^\bullet, \mu_n^\bullet)$ automatically: the norm of the restriction is $\sum_{v \in V_n^\bullet} |f(v)|^2$, which is at most $\sum_{w \in V_{n+1}^\bullet} |f(w)|^2 = \|f\|_{X_{n+1}^0}^2$ since the smaller sum has fewer terms (all nonneg). Probability normalization would introduce a level-dependent rescaling factor that breaks this property.

Notation 3.3 (Sector fibres). We import from [2, §§3, 5, 6, 7]:

- $V_{H_4} \cong \mathbb{H} \cong \mathbb{R}^4$ with the standard Euclidean inner product (square-norm of unit icosian = 1);
- $V_{D_4} \cong \mathbb{R}^4$ with the standard Euclidean inner product;
- $V_{16} = \text{Cl}(1, 3)$ with the Hilbert–Schmidt-style trace form $\langle a, b \rangle_{\text{Cl}} := \langle a^\dagger b \rangle_0$ (positive-definite by [9, Ch. 9]);
- $V_8 = \mathbb{O}$ with the standard octonion norm form $\langle a, b \rangle_{\mathbb{O}} := \Re(a\bar{b})$.

We fix the combined vertex fibre

$$F^0 := V_{H_4} \oplus V_{D_4} \oplus V_{16} \oplus V_8,$$

$\mathbb{R}$ -dimension $4 + 4 + 16 + 8 = 32$, with the orthogonal-direct-sum inner product, and the edge fibre

$$F^1 := \text{Im}(\mathbb{O}),$$

$\mathbb{R}$ -dimension 7, with the Euclidean inner product induced from $\mathbb{O}$. These choices are fixed throughout this paper. (Downstream papers needing a different edge fibre re-construct the relevant spaces from scratch; we make no template promise to accommodate arbitrary F^1 .)

4. SECTOR FIBRES AND FINITE-LEVEL ℓ^2 SPACES

Definition 4.1 (Vertex-sector ℓ^2 space). For $n \geq 0$ and $\bullet \in \{H_4, D_4\}$, define

$$X_n^{0,\bullet} := \ell^2(V(G_n^\bullet), \mu_n^\bullet; F^0),$$

the F^0 -valued ℓ^2 -functions on $V(G_n^\bullet)$ under counting measure, with inner product

$$\langle f, g \rangle_{X_n^0} := \sum_{v \in V(G_n^\bullet)} \langle f(v), g(v) \rangle_{F^0}.$$

Definition 4.2 (Edge-sector ℓ^2 space). For $n \geq 0$ define

$$X_n^{1,\bullet} := \ell_{\text{anti}}^2(E^{\text{or}}(G_n^\bullet), \nu_n^\bullet; F^1),$$

the F^1 -valued ℓ^2 -functions A on oriented edges satisfying the antisymmetry convention $A(-e) = -A(e)$ for every oriented edge e (where $-e$ denotes reversed orientation), with inner product

$$\langle A, B \rangle_{X_n^1} := \frac{1}{2} \sum_{e \in E^{\text{or}}(G_n^\bullet)} \langle A(e), B(e) \rangle_{F^1}.$$

The factor of $\frac{1}{2}$ accounts for the double-counting from including both orientations of each unoriented edge.

Proposition 4.3 (Finite-level Hilbert structure). *For every $n \geq 0$ and $\bullet \in \{H_4, D_4\}$:*

- (1) $X_n^{0,\bullet}$ and $X_n^{1,\bullet}$ are finite-dimensional real Hilbert spaces.
- (2) $\dim_{\mathbb{R}} X_n^{0,\bullet} = |V(G_n^\bullet)| \cdot 32$ and $\dim_{\mathbb{R}} X_n^{1,\bullet} = |E(G_n^\bullet)| \cdot 7$, where $|E(G_n^\bullet)|$ is the number of unoriented edges ($= |E^{\text{or}}(G_n^\bullet)|/2$).
- (3) In particular, both spaces are separable Hilbert spaces.

Proof. Each fibre is a finite-dimensional Euclidean space (32 for F^0 , 7 for F^1), and $V(G_n^\bullet), E^{\text{or}}(G_n^\bullet)$ are finite (Proposition 2.8). A finite direct sum of finite-dimensional Euclidean spaces under a positive-definite bilinear form is a finite-dimensional Hilbert space [4, Ch. II]. The dimension formulas follow by counting; the antisymmetry convention identifies each pair $(e, -e)$ with a single element of dimension $\dim_{\mathbb{R}} F^1 = 7$, giving the unoriented-edge count. Separability is automatic. $\square$

5. DOWNWARD BONDING MAPS AND THEIR ADJOINTS

5.1. Vertex-sector bonding by restriction.

Definition 5.1 (Vertex bonding map). For $n \geq 0$, define the *vertex bonding map*

$$p_{n+1,n}^0: X_{n+1}^0 \rightarrow X_n^0, \quad (p_{n+1,n}^0 f)(v) := f(v) \quad \text{for } v \in V(G_n^\bullet).$$

Since $V(G_n^\bullet) \subset V(G_{n+1}^\bullet)$ (Definition 2.3), the right-hand side is well-defined: f is already a function on $V(G_{n+1}^\bullet)$ and we simply restrict its domain to the subset $V(G_n^\bullet)$.

5.2. Edge-sector bonding by parent-fibre averaging.

Definition 5.2 (Edge bonding map). For $n \geq 0$, define the *edge bonding map*

$$p_{n+1,n}^1: X_{n+1}^1 \rightarrow X_n^1, \quad (p_{n+1,n}^1 A)(e) := \frac{1}{2} \sum_{e' \in (\pi_{n+1,n}^E)^{-1}(e)} A(e')$$

for $e \in E^{\text{or}}(G_n^\bullet)$. By Proposition 2.8(4), the parent fibre $(\pi_{n+1,n}^E)^{-1}(e)$ has exactly two oriented edges (the two subdividing pieces of e), so the sum has two terms and the average factor $\frac{1}{2}$ gives the standard arithmetic mean.

We verify that $p_{n+1,n}^1 A$ satisfies the antisymmetry convention: $(p_{n+1,n}^1 A)(-e) = \frac{1}{2} \sum A(-e') = -\frac{1}{2} \sum A(e') = -(p_{n+1,n}^1 A)(e)$, since each subdividing piece of $-e$ is the reverse of a subdividing piece of e .

5.3. Bonding maps are contractions.

Theorem 5.3 (Bonding contractions). *For every $n \geq 0$:*

- (1) $p_{n+1,n}^0: X_{n+1}^0 \rightarrow X_n^0$ is a contraction: $\|p_{n+1,n}^0 f\|_{X_n^0} \leq \|f\|_{X_{n+1}^0}$ for all $f \in X_{n+1}^0$.
- (2) $p_{n+1,n}^1: X_{n+1}^1 \rightarrow X_n^1$ is a contraction: $\|p_{n+1,n}^1 A\|_{X_n^1} \leq \|A\|_{X_{n+1}^1}$ for all $A \in X_{n+1}^1$.

Proof. (1) By Definition 5.1, $\|p_{n+1,n}^0 f\|_{X_n^0}^2 = \sum_{v \in V(G_n^\bullet)} \|f(v)\|_{F^0}^2$. Since $V(G_n^\bullet) \subset V(G_{n+1}^\bullet)$ and $\|f(v)\|_{F^0}^2 \geq 0$ for every v ,

$$\|p_{n+1,n}^0 f\|_{X_n^0}^2 = \sum_{v \in V(G_n^\bullet)} \|f(v)\|_{F^0}^2 \leq \sum_{w \in V(G_{n+1}^\bullet)} \|f(w)\|_{F^0}^2 = \|f\|_{X_{n+1}^0}^2.$$

(2) By Definition 5.2 and the Cauchy–Schwarz inequality applied to the two-term average $\frac{1}{2}(A(e_1) + A(e_2))$:

$$\frac{1}{2} \|(p_{n+1,n}^1 A)(e)\|_{F^1}^2 = \frac{1}{2} \cdot \frac{1}{4} \|A(e_1) + A(e_2)\|_{F^1}^2 \leq \frac{1}{2} \cdot \frac{1}{2} (\|A(e_1)\|_{F^1}^2 + \|A(e_2)\|_{F^1}^2),$$

where the first equality is the antisymmetry convention's $\frac{1}{2}$ factor and the inequality is Cauchy–Schwarz on the two-term average. Summing over $e \in E^{\text{or}}(G_n^\bullet)$ and using that each subdividing piece e' of level $n+1$ contributes to exactly one parent (by Proposition 2.8(4)):

$$\|p_{n+1,n}^1 A\|_{X_n^1}^2 \leq \frac{1}{2} \sum_{e' \in E_{\text{sub}}^{\text{or}}(G_{n+1}^\bullet)} \|A(e')\|_{F^1}^2 \leq \frac{1}{2} \sum_{e' \in E^{\text{or}}(G_{n+1}^\bullet)} \|A(e')\|_{F^1}^2 = \|A\|_{X_{n+1}^1}^2,$$

where the final inequality is because $E_{\text{sub}}^{\text{or}}(G_{n+1}^\bullet) \subseteq E^{\text{or}}(G_{n+1}^\bullet)$. □

5.4. Adjoint cylindrical embeddings.

Definition 5.4 (Cylindrical embeddings). For $n \geq 0$, define:

- *Vertex embedding* $j_{n,n+1}^0: X_n^0 \rightarrow X_{n+1}^0$ by zero-extension:

$$(j_{n,n+1}^0 g)(w) := \begin{cases} g(w) & \text{if } w \in V(G_n^\bullet), \\ 0 & \text{if } w \in V(G_{n+1}^\bullet) \setminus V(G_n^\bullet). \end{cases}$$

- *Edge embedding (adjoint of p^1)* $j_{n,n+1}^1: X_n^1 \rightarrow X_{n+1}^1$ by rescaled parent-replication:

$$(j_{n,n+1}^1 B)(e') := \begin{cases} \frac{1}{2} B(\pi_{n+1,n}^E(e')) & \text{if } \pi_{n+1,n}^E(e') \neq \perp, \\ 0 & \text{otherwise.} \end{cases}$$

The rescaling factor $\frac{1}{2}$ is required for $j_{n,n+1}^1$ to be the genuine ℓ^2 -adjoint of $p_{n+1,n}^1$ (Proposition 5.5(3) below). Without this factor, the unscaled replication $s_{n,n+1}^1 B(e') := B(\pi_{n+1,n}^E(e'))$ is a *section* of $p_{n+1,n}^1$ (i.e. $p_{n+1,n}^1 \circ s_{n,n+1}^1 = \text{id}$, Remark 5.6 below) but not its adjoint.

Proposition 5.5 (Adjoint relation). *For every $n \geq 0$:*

- (1) $j_{n,n+1}^0 = (p_{n+1,n}^0)^*$ as bounded linear maps, i.e. $\langle p_{n+1,n}^0 f, g \rangle_{X_n^0} = \langle f, j_{n,n+1}^0 g \rangle_{X_{n+1}^0}$ for all $f \in X_{n+1}^0, g \in X_n^0$.
- (2) $p_{n+1,n}^0 \circ j_{n,n+1}^0 = \text{id}_{X_n^0}$ (vertex embedding is a section).
- (3) $j_{n,n+1}^1 = (p_{n+1,n}^1)^*$ as bounded linear maps.
- (4) $p_{n+1,n}^1 \circ j_{n,n+1}^1 = \frac{1}{2} \text{id}_{X_n^1}$ (the genuine adjoint j^1 is a half-section, not an exact section, of p^1 ; see also Remark 5.6).

Proof. (1)

$$\begin{aligned} \langle p_{n+1,n}^0 f, g \rangle_{X_n^0} &= \sum_{v \in V(G_n^\bullet)} \langle f(v), g(v) \rangle_{F^0} \\ &= \sum_{w \in V(G_{n+1}^\bullet)} \langle f(w), (j_{n,n+1}^0 g)(w) \rangle_{F^0} \quad (\text{extending by zero}) \\ &= \langle f, j_{n,n+1}^0 g \rangle_{X_{n+1}^0}. \end{aligned}$$

(2) For $g \in X_n^0$ and $v \in V(G_n^\bullet)$, $(p_{n+1,n}^0 j_{n,n+1}^0 g)(v) = (j_{n,n+1}^0 g)(v) = g(v)$.

(3) Compute both sides explicitly. With the edge-sector inner product factor of $\frac{1}{2}$ from Definition 4.2:

$$\begin{aligned} \langle p_{n+1,n}^1 A, B \rangle_{X_n^1} &= \frac{1}{2} \sum_{e \in E^{\text{or}}(G_n^\bullet)} \langle (p_{n+1,n}^1 A)(e), B(e) \rangle_{F^1} \\ &= \frac{1}{2} \sum_e \frac{1}{2} \sum_{e' \in (\pi^E)^{-1}(e)} \langle A(e'), B(e) \rangle_{F^1} \\ &= \frac{1}{4} \sum_{e' \in E_{\text{sub}}^{\text{or}}(G_{n+1}^\bullet)} \langle A(e'), B(\pi^E(e')) \rangle_{F^1}. \end{aligned}$$

With the rescaled $j_{n,n+1}^1$ formula $(j_{n,n+1}^1 B)(e') = \frac{1}{2} B(\pi^E(e'))$ on subdividing edges and 0 elsewhere:

$$\begin{aligned} \langle A, j_{n,n+1}^1 B \rangle_{X_{n+1}^1} &= \frac{1}{2} \sum_{e' \in E^{\text{or}}(G_{n+1}^\bullet)} \langle A(e'), (j_{n,n+1}^1 B)(e') \rangle_{F^1} \\ &= \frac{1}{2} \sum_{e' \in E_{\text{sub}}^{\text{or}}} \langle A(e'), \frac{1}{2} B(\pi^E(e')) \rangle_{F^1} \\ &= \frac{1}{4} \sum_{e' \in E_{\text{sub}}^{\text{or}}} \langle A(e'), B(\pi^E(e')) \rangle_{F^1}. \end{aligned}$$

The two expressions agree, so $j_{n,n+1}^1 = (p_{n+1,n}^1)^*$.

(4) For $B \in X_n^1$ and $e \in E^{\text{or}}(G_n^\bullet)$, the parent fibre of e has two oriented edges e_1, e_2 , both satisfying $\pi^E(e_i) = e$. Thus $(j_{n,n+1}^1 B)(e_i) = \frac{1}{2} B(e)$ for $i = 1, 2$, and $(p_{n+1,n}^1 j_{n,n+1}^1 B)(e) = \frac{1}{2} ((j_{n,n+1}^1 B)(e_1) + (j_{n,n+1}^1 B)(e_2)) = \frac{1}{2} (\frac{1}{2} B(e) + \frac{1}{2} B(e)) = \frac{1}{2} B(e)$. So $p_{n+1,n}^1 \circ j_{n,n+1}^1 = \frac{1}{2} \text{id}_{X_n^1}$. $\square$

Remark 5.6 (Edge section vs adjoint). The edge sector has *two* naturally-related upward maps from X_n^1 to X_{n+1}^1 :

- the *adjoint* $j_{n,n+1}^1 := (p_{n+1,n}^1)^*$, which satisfies $j_{n,n+1}^1 B(e') = \frac{1}{2} B(\pi^E(e'))$ on subdividing edges and $p_{n+1,n}^1 \circ j_{n,n+1}^1 = \frac{1}{2} \text{id}$ (Proposition 5.5(4));
- the *section* $s_{n,n+1}^1 B(e') := B(\pi^E(e'))$ on subdividing edges and 0 elsewhere, which satisfies $p_{n+1,n}^1 \circ s_{n,n+1}^1 = \text{id}$ but is not the ℓ^2 -adjoint of $p_{n+1,n}^1$.

The two differ by the rescaling $j_{n,n+1}^1 = \frac{1}{2} s_{n,n+1}^1$. We use j (the adjoint) below for inverse-limit constructions; we use s (the section) only when an exact section is needed downstream. The vertex sector does not have this distinction because the vertex bonding map is restriction (not averaging), and the corresponding adjoint $j_{n,n+1}^0$ is zero-extension, which is a section ($p^0 \circ j^0 = \text{id}$).

6. THE CYLINDRICAL ALGEBRAIC DIRECT-LIMIT SPACE

6.1. Algebraic direct limit.

Definition 6.1 (Algebraic direct limit on vertices). The *cylindrical algebraic direct-limit space* (vertex sector) is

$$X_{\text{cyl}}^{0,\bullet} := \varinjlim_n (X_n^{0,\bullet}, j_{n,n+1}^0),$$

the direct limit of the system $(X_n^{0,\bullet})_{n \geq 0}$ under the cylindrical embeddings $j_{n,n+1}^0$ of Definition 5.4.

Theorem 6.2 (Identification of $X_{\text{cyl}}^{0,\bullet}$). *The vertex cylindrical direct-limit space identifies canonically with c_{00} -functions on the countable union $V_\infty^\bullet$ of refinement-vertex sets:*

$$X_{\text{cyl}}^{0,\bullet} \cong c_{00}(V_\infty^\bullet; F^0),$$

where $V_\infty^\bullet := \bigcup_{n \geq 0} V(G_n^\bullet)$ (a countable set, by Proposition 2.8(1) and the inclusion chain) and $c_{00}(\cdot; F^0)$ denotes F^0 -valued functions of finite support.

Proof. The algebraic direct limit $\varinjlim_n X_n^{0,\bullet}$ is the disjoint union $\bigsqcup_n X_n^{0,\bullet}$ modulo the equivalence relation generated by $f \sim j_{n,n+1}^0 f$ for $f \in X_n^{0,\bullet}$ [4, Ch. II.1.4]. Each equivalence class has a unique representative at the smallest level n for which the class contains an element; that representative is a function on $V(G_n^\bullet)$, which by zero-extension to all higher levels gives a function on $V_\infty^\bullet$ vanishing outside $V(G_n^\bullet)$.

Conversely, every c_{00} -function $\phi: V_\infty^\bullet \rightarrow F^0$ has support contained in $V(G_n^\bullet)$ for some n (since the support is finite and $V_\infty^\bullet$ is the ascending union), so it represents an element of $X_n^{0,\bullet}$.

These maps are inverse to each other and are $\mathbb{R}$ -linear, giving the identification $X_{\text{cyl}}^{0,\bullet} \cong c_{00}(V_\infty^\bullet; F^0)$. $\square$

Remark 6.3 (Edge sector: half-section, finite-level only). The edge cylindrical direct-limit construction of the vertex sector does not transfer cleanly to edges. The adjoint $j_{n,n+1}^1 = (p_{n+1,n}^1)^*$ satisfies only $p_{n+1,n}^1 \circ j_{n,n+1}^1 = \frac{1}{2} \text{id}_{X_n^1}$ (Proposition 5.5(4)): it is a *half-section*, not a section, of $p_{n+1,n}^1$. Consequently $\varinjlim_n (X_n^{1,\bullet}, j_{n,n+1}^1)$ does not enjoy the properties used in the vertex identification (Theorem 6.2), and we do *not* define an edge cylindrical direct-limit space here. Downstream papers requiring the edge sector should work at the finite level X_n^1 with the bounded bonding maps $p_{n+1,n}^1$ and their adjoints, without relying on a cylindrical-direct-limit formalism for edges.

 6.2. The ℓ^2 -norm on X_{cyl} .

Definition 6.4 (Pre-Hilbert structure on X_{cyl}). On $X_{\text{cyl}}^{0,\bullet} \cong c_{00}(V_\infty^\bullet; F^0)$, define the inner product

$$\langle \phi, \psi \rangle_{X_{\text{cyl}}^0} := \sum_{v \in V_\infty^\bullet} \langle \phi(v), \psi(v) \rangle_{F^0}.$$

This sum is finite since ϕ, ψ have finite support. The resulting inner product is positive-definite (since $\langle \cdot, \cdot \rangle_{F^0}$ is) and makes X_{cyl}^0 into a pre-Hilbert space.

Lemma 6.5 (Compatibility with finite-level inner products). *For each $n \geq 0$ and $f \in X_n^{0,\bullet}$ identified with its image in $X_{\text{cyl}}^{0,\bullet}$ via the embedding $j_{n,\infty}^0$ (zero-extension to $V_\infty^\bullet$), $\|f\|_{X_{\text{cyl}}^0} = \|f\|_{X_n^0}$.*

Proof. $\|f\|_{X_n^0}^2 = \sum_{v \in V(G_n^\bullet)} \|f(v)\|_{F^0}^2$; $\|f\|_{X_{\text{cyl}}^0}^2 = \sum_{v \in V_\infty^\bullet} \|f(v)\|^2$, where $f(v) = 0$ for $v \in V_\infty^\bullet \setminus V(G_n^\bullet)$. The two sums agree. $\square$

7. THE INVERSE-LIMIT HILBERT SPACE

Definition 7.1 (Inverse-limit space). The *vertex inverse-limit space* is

$$X_{\text{inv}}^{0,\bullet} := \{(f_n)_{n \geq 0} : f_n \in X_n^{0,\bullet}, p_{n+1,n}^0 f_{n+1} = f_n \text{ for all } n, \sup_n \|f_n\|_{X_n^0} < \infty\}.$$

It is equipped with the inner product

$$\langle f, g \rangle_{X_{\text{inv}}^0} := \lim_{n \rightarrow \infty} \langle f_n, g_n \rangle_{X_n^0}.$$

Lemma 7.2 (Limits exist). *For every bounded compatible family $(f_n)_n \in X_{\text{inv}}^{0,\bullet}$, the sequence $\|f_n\|_{X_n^0}$ is non-decreasing and bounded, hence converges. By polarization, $\langle f_n, g_n \rangle_{X_n^0}$ also converges for any pair of compatible families $(f_n), (g_n)$.*

Proof. By Definition 5.1, $p_{n+1,n}^0$ is restriction to $V(G_n^\bullet) \subset V(G_{n+1}^\bullet)$. So the compatibility $f_n = p_{n+1,n}^0 f_{n+1}$ means f_n is the restriction of f_{n+1} to the smaller domain $V(G_n^\bullet)$. Hence $\|f_n\|^2 = \sum_{v \in V(G_n^\bullet)} \|f_{n+1}(v)\|^2 \leq \sum_{w \in V(G_{n+1}^\bullet)} \|f_{n+1}(w)\|^2 = \|f_{n+1}\|^2$. The sequence is non-decreasing. By the supremum bound it is bounded, hence converges. Polarization for cross terms: $\langle f_n, g_n \rangle = \frac{1}{4}(\|f_n + g_n\|^2 - \|f_n - g_n\|^2)$, both norms converge, so does the inner product. $\square$

Theorem 7.3 ($X_{\text{inv}}^0 \cong \ell^2(V_\infty^\bullet)$). *The map*

$$\Phi: X_{\text{inv}}^{0,\bullet} \rightarrow \ell^2(V_\infty^\bullet; F^0), \quad \Phi((f_n)_n) := f_\infty,$$

where $f_\infty(v) := f_n(v)$ for $v \in V(G_n^\bullet)$ (using the compatibility $f_n = p_{n+1,n}^0 f_{n+1}$ to ensure well-definedness across levels), is an $\mathbb{R}$ -linear bijection that intertwines the inner products:

$$\langle f, g \rangle_{X_{\text{inv}}^0} = \langle \Phi(f), \Phi(g) \rangle_{\ell^2(V_\infty^\bullet; F^0)}.$$

In particular, $X_{\text{inv}}^{0,\bullet}$ is a separable Hilbert space.

Proof. *Well-definedness of Φ .* For $v \in V_\infty^\bullet = \bigcup_n V(G_n^\bullet)$, choose n minimal such that $v \in V(G_n^\bullet)$ and set $f_\infty(v) := f_n(v)$. The compatibility $f_n = p_{n+1,n}^0 f_{n+1}$ gives $f_n(v) = f_{n+1}(v)$ for $v \in V(G_n^\bullet)$ at any larger level, so the value is independent of the choice of n .

$f_\infty \in \ell^2(V_\infty^\bullet; F^0)$: by Lemma 7.2, $\sum_v \|f_\infty(v)\|^2 = \lim_n \sum_{v \in V(G_n^\bullet)} \|f_n(v)\|^2 = \lim_n \|f_n\|^2 < \infty$.

Injectivity. If $\Phi(f) = 0$, then $f_\infty(v) = 0$ for all v , so $f_n(v) = 0$ for all $n, v \in V(G_n^\bullet)$, hence $f_n = 0$ for all n .

Surjectivity. For $\phi \in \ell^2(V_\infty^\bullet; F^0)$, define $f_n := \phi|_{V(G_n^\bullet)} \in X_n^0$. Then $\|f_n\|^2 = \sum_{v \in V(G_n^\bullet)} \|\phi(v)\|^2 \leq \|\phi\|_{\ell^2}^2$, bounded. Compatibility: $p_{n+1,n}^0 f_{n+1} = (\phi|_{V(G_{n+1}^\bullet)})|_{V(G_n^\bullet)} = \phi|_{V(G_n^\bullet)} = f_n$. So $(f_n) \in X_{\text{inv}}$ and $\Phi(f) = \phi$.

Inner product preservation. $\langle f, g \rangle_{X_{\text{inv}}} = \lim_n \sum_{v \in V(G_n^\bullet)} \langle f_n(v), g_n(v) \rangle = \sum_{v \in V_\infty^\bullet} \langle \Phi(f)(v), \Phi(g)(v) \rangle = \langle \Phi(f), \Phi(g) \rangle_{\ell^2}$.

Hilbert + separable. ℓ^2 on a countable set with finite-dimensional fibre is a standard separable Hilbert space [4, Ch. II.4]. The bijection Φ transports this structure to $X_{\text{inv}}^{0,\bullet}$. $\square$

7.1. Density of X_{cyl} in X_{inv} .

Proposition 7.4 (X_{cyl}^0 is dense in X_{inv}^0). *Under the canonical map*

$$X_{\text{cyl}}^{0,\bullet} \cong c_{00}(V_\infty^\bullet; F^0) \hookrightarrow \ell^2(V_\infty^\bullet; F^0) \cong X_{\text{inv}}^{0,\bullet},$$

the cylindrical algebraic direct-limit space embeds as a dense subspace of the inverse-limit Hilbert space.

Proof. c_{00} is dense in ℓ^2 on any countable index set, by truncation: for $\phi \in \ell^2(V_\infty^\bullet)$ and $\varepsilon > 0$, choose n large enough that $\sum_{v \in V_\infty^\bullet \setminus V(G_n^\bullet)} \|\phi(v)\|^2 < \varepsilon^2$, then $\phi_n := \phi|_{V(G_n^\bullet)}$ extended by zero is in c_{00} and satisfies $\|\phi - \phi_n\|^2 < \varepsilon^2$. By Theorem 7.3, the same density holds in $X_{\text{inv}}^{0,\bullet}$. $\square$

7.2. Edge sector.

Remark 7.5 (Edge inverse limit: deferred). The edge bonding map $p_{n+1,n}^1$ is parent-fibre averaging (not restriction), and satisfies the contraction bound $\|p_{n+1,n}^1\| \leq 1$ (Theorem 5.3(2)). However, contraction-monotonicity alone does *not* establish completeness of the set $X_{\text{inv}}^{1,\bullet} := \{(A_n) : p_{n+1,n}^1 A_{n+1} = A_n, \sup_n \|A_n\| < \infty\}$ in the induced limit norm: the vertex proof of Theorem 7.3 uses the concrete identification with $\ell^2(V_\infty^\bullet; F^0)$ (via restriction-bonding maps), and the averaging-bonding edge sector has no analogous identification at present. We therefore do *not* claim here that $X_{\text{inv}}^{1,\bullet}$ is a Hilbert space; it is at this stage a set of bounded compatible families with a pre-inner product induced by the finite-level norms. Establishing Hilbert-completion for the edge sector is left to downstream papers needing it.

8. COMMUTING SYMMETRY ACTIONS

8.1. The three actions on X_n .

Definition 8.1 (Coxeter action, branchwise). For $g \in W^\bullet$ and $f \in X_n^{0,\bullet}$, define $(g \cdot f)(v) := \rho_{F^0}^\bullet(g) \cdot f(g^{-1} \cdot v)$ for $v \in V(G_n^\bullet)$, where the fibre representation $\rho_{F^0}^\bullet$ depends on the branch:

- On the H_4 branch ($\bullet = H_4$, $W^\bullet = W(H_4)$): $\rho_{F^0}^{H_4}(g)$ acts by the standard H_4 -reflection representation on V_{H_4} , and trivially on V_{D_4}, V_{16}, V_8 .
- On the D_4 branch ($\bullet = D_4$, $W^\bullet = W(D_4)$): $\rho_{F^0}^{D_4}(g)$ acts by the standard D_4 -reflection representation on V_{D_4} , and trivially on V_{H_4}, V_{16}, V_8 .

The analogous action on $X_n^{1,\bullet}$ acts trivially on $F^1 = \text{Im}(\mathbb{O})$ on both branches and on the edge index by $W^\bullet$ -permutation.

Lemma 8.2 (Unitarity of the Coxeter action). *Under the ℓ^2 -inner product of Definition 4.1, the Coxeter action of Definition 8.1 is unitary on $X_n^{0,\bullet}$ and on $X_n^{1,\bullet}$.*

Proof. The H_4 - (resp. D_4 -) reflection representation on V_{H_4} (resp. V_{D_4}) is orthogonal with respect to the Euclidean metric on the fibre (P2 §§3–5), and acts trivially on the other fibre summands; hence $\rho_{F^0}^\bullet(g)$ is a fibre isometry. The vertex (resp. oriented-edge) measure $\mu_n^\bullet$ (resp. $\nu_n^\bullet$) is $W^\bullet$ -invariant by the Coxeter-equivariance of the refinement (Proposition 2.8). The combined pre-composition with g^{-1} and fibrewise action by $\rho_{F^0}^\bullet(g)$ is therefore a unitary on the ℓ^2 -space. $\square$

8.2. Q-form and the σ -action.

Definition 8.3 (Mixed $\mathbb{Q}(\varphi)/\mathbb{Q}$ -form of the vertex sector). By inspection of the explicit fibre constructions in [2, §§3, 5, 6, 7]: $V_{D_4} = \mathbb{R}^4$ has the standard $\mathbb{Q}$ -form $V_{D_4,\mathbb{Q}} := \mathbb{Q}^4$; $V_{16} = \text{Cl}(1, 3)$ has the $\mathbb{Q}$ -form $V_{16,\mathbb{Q}} := \mathbb{Q}\{\gamma_A\}$ (rational span of the Clifford monomial basis, [2, Def. 6.2], which is a $\mathbb{Q}$ -basis of a $\mathbb{Q}$ -form whose $\mathbb{R}$ -extension equals $\text{Cl}(1, 3)$); $V_8 = \mathbb{O}$ has the $\mathbb{Q}$ -form $V_{8,\mathbb{Q}} := \mathbb{Q}\{1, e_1, \dots, e_7\}$ (rational span of the standard octonion basis, [2, Def. 7.1]). For $V_{H_4} = \mathbb{R}^4$ the natural rational structure is the $\mathbb{Q}(\varphi)$ -form $V_{H_4,\mathbb{Q}(\varphi)} := \mathbb{Q}(\varphi)^4 \subset \mathbb{R}^4$ via the standard real embedding $\iota: \mathbb{Q}(\varphi) \hookrightarrow \mathbb{R}$ sending φ to the golden ratio $\frac{1}{2}(1 + \sqrt{5})$, because the icosian arithmetic on V_{H_4} is naturally over $\mathbb{Z}[\varphi]$ rather than over $\mathbb{Z}$ [2, §3].

The combined rationality data is the *mixed* $\mathbb{Q}(\varphi)/\mathbb{Q}$ -form

$$F_{\mathbb{Q}(\varphi)/\mathbb{Q}}^0 := V_{H_4,\mathbb{Q}(\varphi)} \oplus V_{D_4,\mathbb{Q}} \oplus V_{16,\mathbb{Q}} \oplus V_{8,\mathbb{Q}},$$

which is a $\mathbb{Q}$ -vector space of total $\mathbb{Q}$ -dimension $8 + 4 + 16 + 8 = 36$ (since $V_{H_4,\mathbb{Q}(\varphi)} = \mathbb{Q}(\varphi)^4$ has $\mathbb{Q}$ -dimension 8). Its $\mathbb{R}$ -extension is sector-dependent: on the H_4 sector we use $V_{H_4,\mathbb{Q}(\varphi)} \otimes_{\mathbb{Q}(\varphi)} \mathbb{R} = \mathbb{R}^4 = V_{H_4}$ via ι , on the other three sectors we use $V_{*,\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{R} = V_*$. With this convention

$$F^0 = V_{H_4} \oplus V_{D_4} \oplus V_{16} \oplus V_8 \quad (\dim_{\mathbb{R}} F^0 = 32)$$

is recovered from $F_{\mathbb{Q}(\varphi)/\mathbb{Q}}^0$. We do *not* claim $F_{\mathbb{Q}(\varphi)/\mathbb{Q}}^0 \otimes_{\mathbb{Q}} \mathbb{R} = F^0$; that would have $\mathbb{R}$ -dimension 36, not 32, because $\mathbb{Q}(\varphi) \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{R} \oplus \mathbb{R}$ (the two real roots of $x^2 - x - 1$). The $\mathbb{R}$ -dimension 32 of F^0 comes from the $\mathbb{Q}(\varphi)$ -tensor product on the H_4 sector, not from the $\mathbb{Q}$ -tensor product.

Define the mixed-form vertex space at level n by

$$X_n^{0,\bullet}|_{\mathbb{Q}(\varphi)/\mathbb{Q}} := \ell_{\text{fin}}^2(V(G_n^\bullet), \mu_n^\bullet; F_{\mathbb{Q}(\varphi)/\mathbb{Q}}^0),$$

the $F_{\mathbb{Q}(\varphi)/\mathbb{Q}}^0$ -valued functions on the finite set $V(G_n^\bullet)$.

Definition 8.4 (σ -action on the mixed-form vertex sector). The Galois involution $\sigma: \mathbb{Q}(\varphi) \rightarrow \mathbb{Q}(\varphi)$ defined by $\sigma(\varphi) = 1 - \varphi$ ([1, Def. def:sigma]) extends sector-by-sector to a $\mathbb{Q}$ -linear involution $\sigma_n: X_n^{0,\bullet}|_{\mathbb{Q}(\varphi)/\mathbb{Q}} \rightarrow X_n^{0,\bullet}|_{\mathbb{Q}(\varphi)/\mathbb{Q}}$ on the mixed-form vertex space of Definition 8.3 as follows:

- on the $V_{H_4, \mathbb{Q}(\varphi)} = \mathbb{Q}(\varphi)^4$ component, σ_n acts coefficientwise by σ in each of the four $\mathbb{Q}(\varphi)$ -coordinates;
- on each of $V_{D_4, \mathbb{Q}}, V_{16, \mathbb{Q}}, V_{8, \mathbb{Q}}$, σ_n acts as the identity (these sectors carry no $\mathbb{Q}(\varphi)$ -rationality).

σ_n is a $\mathbb{Q}$ -linear involution on the $\mathbb{Q}$ -vector space $X_n^{0,\bullet}|_{\mathbb{Q}(\varphi)/\mathbb{Q}}$. It does *not* extend to an $\mathbb{R}$ -linear involution on the $\mathbb{R}$ -extension $X_n^{0,\bullet} = \ell^2(V(G_n^\bullet), \mu_n^\bullet; F^0)$ through the standard real embedding ι (because σ on $\mathbb{Q}(\varphi) \otimes_{\mathbb{Q}(\varphi), \iota} \mathbb{R} = \mathbb{R}$ is fixed by ι but free over the conjugate embedding $\bar{\iota}$ sending φ to $1 - \varphi$); σ_n is the *rationality-side* involution, used in the intertwining identities below at the mixed-form level.

We do not define a σ -action on the edge sector X_n^1 in this paper: the edge fibre $\text{Im}(\mathbb{O})$ has no canonical $\mathbb{Q}(\varphi)$ -coefficient structure relevant to the cascade arithmetic, and downstream papers needing edge σ -actions construct them separately.

8.3. The intertwining theorem.

Theorem 8.5 (Intertwining identities). *For every $n \geq 0$ and $\bullet \in \{H_4, D_4\}$:*

- (1) (*Refinement-Coxeter, vertex sector.*) *For every $g \in W^\bullet$,*

$$g \circ p_{n+1, n}^0 = p_{n+1, n}^0 \circ g \quad \text{on } X_{n+1}^{0,\bullet},$$

and analogously $g \circ j_{n, n+1}^0 = j_{n, n+1}^0 \circ g$.

- (2) (*Refinement-Coxeter, edge sector.*) *The same identities hold for $p_{n+1, n}^1$ and $j_{n, n+1}^1$ on $X_{n+1}^{1,\bullet}$.*

- (3) (*Refinement- σ , vertex sector, mixed-form level.*) *On the mixed-form vertex space $X_{n+1}^{0,\bullet}|_{\mathbb{Q}(\varphi)/\mathbb{Q}}$ of Definition 8.3,*

$$\sigma_n \circ p_{n+1, n}^0|_{\mathbb{Q}(\varphi)/\mathbb{Q}} = p_{n+1, n}^0|_{\mathbb{Q}(\varphi)/\mathbb{Q}} \circ \sigma_{n+1},$$

where $p_{n+1, n}^0|_{\mathbb{Q}(\varphi)/\mathbb{Q}}$ is the mixed-form restriction of the bonding map (well-defined because vertex restriction preserves the sectorwise rationality) and σ_n, σ_{n+1} are the involutions of Definition 8.4. Analogously $\sigma_n \circ j_{n, n+1}^0|_{\mathbb{Q}(\varphi)/\mathbb{Q}} = j_{n, n+1}^0|_{\mathbb{Q}(\varphi)/\mathbb{Q}} \circ \sigma_{n+1}$.

- (4) (*Sectorwise Coxeter- σ , vertex sector.*) *On the D_4 branch ($\bullet = D_4$, $W^\bullet = W(D_4)$), the action of Definition 8.1 acts on V_{D_4} and trivially on V_{H_4}, V_{16}, V_8 ; since σ acts trivially on $V_{D_4, \mathbb{Q}}, V_{16, \mathbb{Q}}, V_{8, \mathbb{Q}}$, the commutation $g|_{\mathbb{Q}(\varphi)} \circ \sigma_n = \sigma_n \circ g|_{\mathbb{Q}(\varphi)}$ holds automatically for all $g \in W(D_4)$ on $X_n^{0, D_4}|_{\mathbb{Q}(\varphi)}$. On the H_4 branch, the analogous automatic-commutation statement holds only on the σ -trivial summands $V_{D_4, \mathbb{Q}}, V_{16, \mathbb{Q}}, V_{8, \mathbb{Q}}$ (on which $W(H_4)$ acts trivially by Definition 8.1); the full statement on the V_{H_4} summand fails (Remark 8.6).*

The Coxeter intertwining of (1)–(2) descend to the vertex cylindrical and inverse-limit spaces $X_{\text{cyl}}^{0,\bullet}, X_{\text{inv}}^{0,\bullet}$ by linearity and continuity (using Theorem 7.3 for the inverse limit). The σ -intertwining of (3) is stated only at the mixed-form level $X_n^{0,\bullet}|_{\mathbb{Q}(\varphi)/\mathbb{Q}}$; we do not claim a σ -action on the $\mathbb{R}$ -Hilbert spaces $X_n^{0,\bullet}, X_{\text{cyl}}^{0,\bullet}, X_{\text{inv}}^{0,\bullet}$, since (per Definition 8.4) σ does not extend $\mathbb{R}$ -linearly through the standard real embedding ι .

Remark 8.6 (Why Coxeter- σ commutation is restricted). [1, Thm. 4.5] requires a $\mathbb{Q}$ -linear endomorphism T of the $\mathbb{Q}$ -vector space V defined over the σ -fixed $\mathbb{Q}$ -form: concretely, T must restrict to $V \otimes 1 \subset V \otimes_{\mathbb{Q}} \mathbb{Q}(\varphi)$. On the $V_{H_4, \mathbb{Q}}$ component, the $W(H_4)$ action is given by

reflection matrices whose coefficients lie in $\mathbb{Q}(\varphi)$ (not in $\mathbb{Q}$), because the icosian roots have $\mathbb{Q}(\varphi)$ -rational coordinates [2, §3]. Such a reflection is $\mathbb{Q}(\varphi)$ -linear (hence a fortiori $\mathbb{Q}$ -linear on the underlying $\mathbb{Q}$ -vector space $V_{H_4, \mathbb{Q}(\varphi)}$), but it does *not* restrict to a $\mathbb{Q}$ -linear endomorphism of the σ -fixed subspace $V_{H_4, \mathbb{Q}} \otimes 1$: a non-trivial $\mathbb{Q}(\varphi)$ -coefficient reflection of a $\mathbb{Q}$ -vector carries it out of $V_{H_4, \mathbb{Q}} \otimes 1$ into the general $V_{H_4, \mathbb{Q}(\varphi)}$ space. The hypothesis of [1, Thm. 4.5] is therefore not satisfied in the form required for automatic σ -commutation, and in general $g \circ \sigma$ and $\sigma \circ g$ do *not* commute on the V_{H_4} sector for $g \in W(H_4)$ acting non-trivially on V_{H_4} .

The sectorwise statement (4) records the cases where commutation *does* hold automatically under the branchwise action of Definition 8.1: the entire D_4 -branch action (since $W(D_4)$ acts only on V_{D_4} , which is σ -trivial), and the H_4 -branch action restricted to the σ -trivial summands $V_{D_4, \mathbb{Q}}, V_{16, \mathbb{Q}}, V_{8, \mathbb{Q}}$ (on which $W(H_4)$ acts trivially).

Downstream papers needing the full $W(H_4)$ - σ relationship (e.g. in spectral-arithmetic constructions) should formulate it as an intertwining with σ -conjugate Coxeter elements g^σ rather than as commutation: $\sigma \circ g = g^\sigma \circ \sigma$ where g^σ is the matrix-coefficient σ -conjugate of g .

Proof. (1) Refinement-Coxeter, vertex. For $f \in X_{n+1}^0, v \in V(G_n^\bullet), g \in W^\bullet$:

$$\begin{aligned} (g \circ p_{n+1, n}^0 f)(v) &= \rho_{F^0}(g) \cdot (p_{n+1, n}^0 f)(g^{-1} \cdot v) = \rho_{F^0}(g) \cdot f(g^{-1} \cdot v) \\ &= (g \cdot f)(v) = (p_{n+1, n}^0 (g \cdot f))(v). \end{aligned}$$

This uses $g^{-1} \cdot v \in V(G_n^\bullet)$ (because $V(G_n^\bullet)$ is $W^\bullet$ -stable) and the equality $f(g^{-1}v) = (p_{n+1, n}^0 f)(g^{-1}v)$ (definition of p^0 as restriction).

The j -version follows by adjunction: $g \circ j = j \circ g$ is equivalent to $g^{-1} \circ p = p \circ g^{-1}$ (which we just proved with g^{-1} in place of g).

(2) *Refinement-Coxeter, edge.* For $A \in X_{n+1}^1, e \in E^{\text{or}}(G_n^\bullet), g \in W^\bullet$ (with the $W^\bullet$ action on $\text{Im}(\mathbb{O})$ trivial, so we drop the $\rho_{F^1}(g)$ factor):

$$(g \circ p_{n+1, n}^1 A)(e) = (p_{n+1, n}^1 A)(g^{-1} \cdot e) = \frac{1}{2} \sum_{e' \in (\pi^E)^{-1}(g^{-1}e)} A(e').$$

By Proposition 2.8(3), $(\pi^E)^{-1}(g^{-1}e) = g^{-1} \cdot (\pi^E)^{-1}(e)$. Substituting $e'' := g \cdot e'$:

$$= \frac{1}{2} \sum_{e'' \in (\pi^E)^{-1}(e)} A(g^{-1} \cdot e'') = \frac{1}{2} \sum_{e''} (g \cdot A)(e'') = (p_{n+1, n}^1 (g \cdot A))(e).$$

(3) *Refinement- σ .* The bonding map $p_{n+1, n}^0$ restricts to a $\mathbb{Q}$ -linear map $p_{n+1, n}^0|_{\mathbb{Q}}: X_{n+1}^{0, \bullet}|_{\mathbb{Q}} \rightarrow X_n^{0, \bullet}|_{\mathbb{Q}}$ between $\mathbb{Q}$ -vector spaces of finite $\mathbb{Q}$ -dimensions $|V(G_{n+1}^\bullet)| \cdot 36$ and $|V(G_n^\bullet)| \cdot 36$ (using the $\mathbb{Q}$ -dimension of $F_{\mathbb{Q}}^0$ from Definition 8.3; the corresponding $\mathbb{R}$ -dimensions of the unrestricted spaces are $|V(\cdot)| \cdot 32$, but the $\mathbb{Q}$ -form has $\mathbb{Q}$ -dimension 36 because of the $\mathbb{Q}(\varphi)$ -coefficient pair on the V_{H_4} sector). By the general functoriality theorem [1, Thm. 4.5], the scalar extension $p_{n+1, n}^0|_{\mathbb{Q}} \otimes \text{id}_{\mathbb{Q}(\varphi)}$ commutes with the coefficientwise σ -actions on both sides. The same argument applies to $j_{n, n+1}^0|_{\mathbb{Q}}$.

(4) *Sectorwise Coxeter- σ .* On the D_4 branch, $W(D_4)$ acts (by Definition 8.1) on V_{D_4} and trivially on V_{H_4}, V_{16}, V_8 ; since σ acts trivially on $V_{D_4, \mathbb{Q}}$ and on $V_{16, \mathbb{Q}}, V_{8, \mathbb{Q}}$, the composition $g \circ \sigma$ equals $\sigma \circ g$ on every fibre summand, so it equals $\sigma \circ g$ on $X_n^{0, D_4}|_{\mathbb{Q}(\varphi)}$. On the H_4 branch, $W(H_4)$ acts on V_{H_4} and trivially on V_{D_4}, V_{16}, V_8 ; on the latter three summands σ also acts trivially, so the commutation is automatic there by the same fibrewise computation. The remaining V_{H_4} summand is excluded from (4) (Remark 8.6).

The descent to X_{cyl} and X_{inv} : each action is continuous on each X_n , and the intertwining is preserved under the algebraic direct-limit construction (Theorem 6.2) and the ℓ^2 -completion (Theorem 7.3). $\square$

9. CITATION CONTRACT AND DOWNSTREAM HANDOFF

9.1. What downstream papers may cite.

(1) Definitions 4.1, 4.2: finite-level Hilbert sector spaces $X_n^{0, \bullet}, X_n^{1, \bullet}$ under counting measure.

- (2) Theorem 5.3: contractive downward bonding maps $p_{n+1,n}^0$ (restriction) and $p_{n+1,n}^1$ (parent-fibre averaging).
- (3) Proposition 5.5: cylindrical embeddings $j_{n,n+1}^0, j_{n,n+1}^1$ as adjoints of the bonding maps, with the section relations $p_{n+1,n}^0 \circ j_{n,n+1}^0 = \text{id}_{X_n^0}$ (vertex sector) and $p_{n+1,n}^1 \circ j_{n,n+1}^1 = \frac{1}{2} \text{id}_{X_n^1}$ (edge sector); see also Remark 5.6 for the unscaled section s^1 in the edge sector.
- (4) Theorem 6.2: $X_{\text{cyl}}^{0,\bullet} \cong c_{00}(V_\infty^\bullet; F^0)$ as algebraic direct limit (vertex sector).
- (5) Theorem 7.3: $X_{\text{inv}}^{0,\bullet} \cong \ell^2(V_\infty^\bullet; F^0)$ as separable Hilbert space (vertex sector).
- (6) Proposition 7.4: $X_{\text{cyl}}^{0,\bullet}$ dense in $X_{\text{inv}}^{0,\bullet}$.
- (7) Theorem 8.5: refinement-Coxeter (full), refinement- σ (full), and Coxeter- σ (restricted, see Remark 8.6) intertwining identities.

9.2. Recommended citation phrasing. “By P3 [3], the cascade vertex substrate is $\ell^2(V_\infty^\bullet; F^0)$ as the projective inverse limit of $(X_n^{0,\bullet}, p_{n+1,n}^0)_n$ with $p_{n+1,n}^0$ restriction (Theorems 5.3, 7.3); the cylindrical direct limit $X_{\text{cyl}}^{0,\bullet} \cong c_{00}(V_\infty^\bullet; F^0)$ embeds densely in $X_{\text{inv}}^{0,\bullet}$ (Theorem 6.2, Proposition 7.4); the refinement-Coxeter and refinement- σ intertwining hold without restriction, and the Coxeter- σ intertwining holds in the restricted sense of Theorem 8.5(4) and Remark 8.6 (full $W(H_4)$ - σ commutation on the V_{H_4} sector fails as a structural matter, because a non-trivial $W(H_4)$ reflection does not preserve the σ -fixed $\mathbb{Q}$ -form $V_{H_4, \mathbb{Q}} \otimes 1$, so the hypothesis of [1, Thm. 4.5] is not satisfied on that summand).”

9.3. What this paper does not provide. No closure-functional / dynamics theorem (P4); no metric, gauge, spectral, Hecke, σ -cohomology, or computation correspondence (P5–P13); no continuum-limit identification of X_{inv} with any classical L^2 space; no physical interpretation.

APPENDIX A. NOTATION AUDIT

Symbol	Meaning	Defined
$\bullet$	branch decoration (H_4 or D_4)	Def. 2.1
V_{600}, V_{24}	base vertex sets (P2)	Def. 2.1
$G_n^\bullet$	graph at refinement level n	Def. 2.3
$V(G_n^\bullet)$	vertex set, with $V_n \subset V_{n+1}$	Def. 2.3
$M_n^\bullet, C_n^\bullet$	midpoints and centroids at level n	Def. 2.3
$\pi_{n+1,n}^E$	canonical edge-parent map	Def. 2.6
$\mu_n^\bullet, \nu_n^\bullet$	vertex / edge counting measures	Def. 3.1
F^0	vertex fibre $V_{H_4} \oplus V_{D_4} \oplus V_{16} \oplus V_8$	Notation 3.3
F^1	edge fibre $\text{Im}(\mathbb{O})$ (fixed)	Notation 3.3
X_n^0, X_n^1	finite-level ℓ^2 spaces	Defs. 4.1, 4.2
$p_{n+1,n}^0$	vertex bonding (restriction)	Def. 5.1
$p_{n+1,n}^1$	edge bonding (parent-fibre averaging)	Def. 5.2
$j_{n,n+1}^0$	vertex embedding (zero-extension)	Def. 5.4
$j_{n,n+1}^1$	edge embedding (parent-replication)	Def. 5.4
$V_\infty^\bullet$	$\bigcup_n V(G_n^\bullet)$	Thm. 6.2
$X_{\text{cyl}}^{0,\bullet}$	$\cong c_{00}(V_\infty^\bullet; F^0)$	Def. 6.1, Thm. 6.2
$X_{\text{inv}}^{0,\bullet}$	$\cong \ell^2(V_\infty^\bullet; F^0)$	Def. 7.1, Thm. 7.3
σ	coefficientwise Galois involution per P1	Def. 8.4
$W^\bullet$	Coxeter group ($W(H_4)$ or $W(D_4)$)	Def. 2.1

REFERENCES

- [1] *Cascade-Classical Correspondence I: Sigma-Fixed Rationality over $\mathbb{Q}(\varphi)$* , papers/cascade-sigma-rationality/cascade-sigma-rationality.tex. (Paper P1.) Pinned results used in this paper: Definition 2.5 (coefficientwise σ_V on $V \otimes_{\mathbb{Q}} \mathbb{Q}(\varphi)$); Definition 4.1 (the projector Π_σ); Lemma 2.6 (σ commutes with $\mathbb{Q}$ -linear endomorphisms after scalar extension); Theorem 4.5 (functoriality of the scalar extension of any $\mathbb{Q}$ -linear map between finite-dimensional $\mathbb{Q}$ -vector spaces).

- [2] *Cascade-Classical Correspondence II: The Cascade Algebraic Substrate*, `papers/cascade-algebraic-substrate/cascade-algebraic-substrate.tex`. (Paper P2.) Pinned results used in this paper: §3 (icosian model of V_{H_4} , $\mathbb{Q}(\varphi)$ -rationality of the $W(H_4)$ matrix entries); §5 (the V_{600} and V_{24} vertex sets, including their explicit coordinates); Proposition 5.3 ($V_{24} \subset V_{600}$ as the rescaled D_4 root polytope); Definition 6.2 (the $\text{Cl}(1, 3)$ monomial basis $\{\gamma_A\}$); Definition 7.1 (the standard octonion basis); §§6, 7 (sector fibre conventions for V_{16} and V_8).
- [3] (This paper.) *Cascade-Classical Correspondence III: Refinement Spaces and Symmetry Actions*.
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